

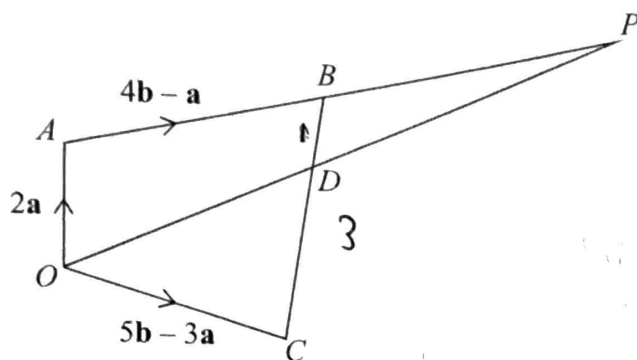


Diagram **NOT**
accurately drawn

$OABC$ is a quadrilateral.
 ABP and ODP are straight lines.

$$\vec{OA} = 2\mathbf{a} \quad \vec{AB} = 4\mathbf{b} - \mathbf{a} \quad \vec{OC} = 5\mathbf{b} - 3\mathbf{a}$$

- (a) Find an expression in terms of $\mathbf{a}$ and $\mathbf{b}$ for the vector $\vec{BC}$.
Simplify your answer.

$$\vec{BC} = \vec{BA} + \vec{AO} + \vec{OC}$$

$$= -(4\mathbf{b} - \mathbf{a}) - 2\mathbf{a} + 5\mathbf{b} - 3\mathbf{a}$$

$$= -4\mathbf{b} + \mathbf{a} - 2\mathbf{a} + 5\mathbf{b} - 3\mathbf{a}$$

$$= 5\mathbf{b} - 4\mathbf{b} - \mathbf{a} - 3\mathbf{a}$$

$$= \underline{\underline{\mathbf{b} - 4\mathbf{a}}}$$

$$\mathbf{b} - 4\mathbf{a}$$

(2)

The point D lies on BC such that $BD:DC = 1:3$

Given that $\vec{OP} = n\vec{OD}$

- (b) use a vector method to find the value of n

